SIGAP - end-to-end test report

27 October 2026 · live SAP S/4HANA and SAP AI Core · lead model aicore:gpt-5.4

What was tested

One disruption handled by the live agent team against real SAP data, then every report the system produces generated from that same record: markdown, PDF, Excel and the operations rollup. A second handling was seeded deterministically, with no model involved, so the reporting path that needs actions, a priced veto and recorded outcomes could be exercised. Both appear in the rollup.

Connectors, live

```
Amazon Bedrock    HTTP 200 in 2.0s - MiniMax M2.5 answered, 74 tokens
SAP AI Core       OAuth ok - 7 deployments - gpt-5.4, sap-rpt-1.6 present
SAP S/4HANA       live - 1,043 stock  2,000 PO  648 sources  1,766 suppliers
Composio          API key read
manage.py check   47 modules ok - 45/45 tools - 12/12 prompts
GET /health       model_ready True - sap_ready True
```

Bedrock answered when called directly. The project refuses to build a Bedrock backend without a second AWS account (core/llm.py _fallback_aws_options), which this bundle does not carry, so the run below used SAP AI Core as the lead.

The agent run

```
event    Supplier halt threatens TG0011 at plant 1010
status   done in 60s - 6 steps - 0 actions - autonomous
```

The team read live SAP and stopped early, on the record: stock on hand 10,020 PC, consumption 0.0 PC/day, no open purchase orders into plant 1010. It stated plainly that no cost figure could be given, and said why it did not call the other specialists: 'Calling more specialists now would not improve the decision enough to justify it.' No action was raised, which is the correct outcome for an exposure it could not prove.

Reporting, generated from that record

```
period                 every handling on record
handlings              2          actions  3
committed, predicted   Rp 123,000,000
outcomes recorded      2  (1 comparable, 1 with no figure)
realized vs predicted  Rp 22,000,000 -> Rp 24,500,000 = +Rp 2,500,000
still owing an outcome 1
what the rules cost    Rp 46,000,000 across 1 disruption
rule pack              0 of 4 verified
```

Checks

Each assertion run against the generated artefacts, not against a mock.

The report says what the rules cost

```
ok   'What the rules cost' section present
ok   Rp 46,000,000 stated
ok   the rule that struck the cheaper plan is named:
     E China spot purchase - TKDN drops to 34.8%, contract floor 40%
```

The report is honest about realized cost

```
ok   realized vs predicted table present
ok   delta Rp 2,500,000 computed deterministically
ok   an outcome with no figure shows '-', never zero
ok   '1 of 2 recorded outcome(s) carry no realized figure' stated
ok   the delta is drawn from comparable actions only, and says so
ok   1 action that went out and still owes an outcome is chased
```

The report is honest about the rules it used

```
ok   '2 of the 2 rules applied here are not verified against a source'
ok   rollup states 0 of 4 rules verified, and names the assumptions
```

Periods

```
ok   ?days=30     -> 'the last 30 days, to 27 Oct 2026 21:10 WIB'
ok   year 2000    -> 0 runs, not everything
ok   mistyped     -> HTTP 400, not widened silently
ok   WIB bounds   -> 1 Oct 00:00 WIB == 30 Sep 17:00 UTC
```

Files

```
ok   PDF and Excel for both handlings and for the rollup
ok   every workbook passes a zip integrity check
ok   the agent run's workbook carries the rows it really read from SAP:
     Summary, Steps, get_open_purchase_orders, get_material_stock
```

The one model-written part

The auxiliary model wrote the executive summary of every report (summary_by: aux). Every figure in its prose traces back to a deterministic one: Rp 123,000,000 is 22 + 94 + 7 million, and it wrote 'for the 1 comparable settled action' rather than claiming the whole book was compared. It introduced no number of its own.

Defect found

Model cost reports Rp 0 on every AI Core run: llm.usd_per_mtok has no price for the aicore provider, so token spend is never priced and SIGAP_COST_CAP_IDR cannot bite while the lead runs there. Invisible in a demo; it does disable the spend cap.

SIGAP

SG SIGAP
connected

Dashboard
Decision queue
Conversation

Agents
Agent team

Records
Data audit
Log
Reports

Settings

R Rina
up to Rp 10 billion

Decision queue

Supply disruptions

2 handlings · 0 need attention

dev · SAP connected · model ready

port closure · done

Typhoon closes Ningbo port

Maritime advisory: CNNGB closed, vessels held a…

Recommend C+B (internal transfer plus Gresik). E blocked on TKDN.

27 Oct 2026 21:15 WIB · no model cost

supplier disruption · done

Supplier halt threatens TG0011 at plant 1010

Supplier notice: production halted, deliveries …

A supplier halt was reported for TG0011 to plant 1010.

27 Oct 2026 21:14 WIB · no model cost · 1 specialist

The decision queue: both handlings, live

dunn port_closure trace

Typhoon closes Ningbo port

Maritime advisory: CNNGB closed, vessels held at anchorage

modelled · e2e-seed

Recommendation

Recommend C+B (internal transfer plus Gresik). E blocked on TKDN.

Not yet trustworthy: compliance.check_local_constraints

Actions · 1 awaiting an outcome

What was sent is only half the record. Close the loop below and the next incident learns from this one.

stock_transfer sent ran on its own authority, under the limit

Material	From plant	To plant
M-4471	SBY1	KRW1
Quantity	**Value**	
22	Rp 22,000,000	

What happened delivered actual Rp 24,500,000 · predicted Rp 22,000,000 (+Rp 2,500,000)

arrived 3 Oct, 22 t

Recorded once on 27 Oct 2026 21:15 WIB by rina@kpn.co.id. A correction is a new reading of reality, not an edit of this one.

draft_po sent ran on its own authority, under the limit

Material	Supplier	Quantity
M-4471	SUP-GRESIK	30
Value		
Rp 94,000,000		

What happened late realized cost not known

arrived 9 Oct, no invoice yet

Recorded once on 27 Oct 2026 21:15 WIB by rina@kpn.co.id. A correction is a new reading of reality, not an edit of this one.

stock_transfer sent ran on its own authority, under the limit

Material	From plant	To plant
M-4471	KRW1	SBY1
Quantity	**Value**	
5	Rp 7,000,000	

This went out. What actually happened?

delivered late short failed cancelled

Actual cost (Rp), if you have the figure: predicted 7000000

What was seen (arrival date, quantity): arrived 3 Oct, 55 t

Leave the cost blank if you do not know it. The report will say the realized cost is not known rather than show a guess, and a figure written into the note stays a note: nothing reads money out of a sentence.

Record what happened

How the system reasoned

Exploration graph | How it reasoned (1)

Data Check
Tara

Comparing scenario costs

▶ Activity details

derived · engine/simulate.py

The disruption page: actions, the outcome control, what was recorded

← Reports

done port_closure

PDF | Excel 5 | Send by email | Markdown | Copy

Typhoon closes Ningbo port

Maritime advisory: CNNGB closed, vessels held at anchorage

Started	Took	Steps	Tool calls	Actions	Model time
27 Oct 2026 21:15 W…	0m 1s	1	1	3	-

Summary

Phrased by the auxiliary model from the figures in this report. It was given no number to invent.

A typhoon closed Ningbo port, with CNNGB closed and vessels held at anchorage, disrupting supply. The decision was to recommend C+B using an internal transfer plus Gresik because E was blocked on TKDN; the rules added Rp 46,000,000 by forcing C, B at Rp 116,000,000 instead of C, E at Rp 70,000,000. Residual risk remains because the local-constraints compliance check is not yet trustworthy, and both TKDN rules used here are unverified assumptions.

Decision

Recommend C+B (Internal transfer plus Gresik). E blocked on TKDN.

Not yet trustworthy: compliance.check_local_constraints

2 of the 2 rules applied here are not verified against a source, so they are assumptions written down rather than rules: tkdn_contract_floor_pct, tkdn_current_pct

What the rules cost

The price of the rule, not an overrun. It is on the record so the decision can be defended without re-deriving it.

Obeying the rules on this disruption cost **Rp 46,000,000**.

Chosen	Cheaper, struck out
C, B	C, E
Rp 116,000,000	Rp 70,000,000

E China spot purchase — TKDN drops to 34.8%, contract floor 40%

How the system reasoned (1 steps)

Open the full trace →

Data Check — Comparing scenario costs

Tara — ▶ Activity details

derived · engine/simulate.py

Assembled from the stored record, not re-summarised by a model, so it says exactly what the log says. The PDF and the workbook are built from the same record.

Actions (3)

stock_transfer	sent	
on its own authority		Rp 22,000,000
delivered		actual Rp 24,500,000
draft_po	sent	
on its own authority		Rp 94,000,000
late		no realized figure
stock_transfer	sent	
on its own authority		Rp 7,000,000
owes an outcome		

In the Excel file (5 sheets)

Summary	17 rows
Compliance cost	6 rows
Actions	3 rows
Steps	1 rows
simulate_scenario	2 rows

Every table the handling produced: its actions and steps, any table in the decision, and each row a tool read, one sheet per tool with the step and source beside it.

Related

The decision page →

The trace →

The report: what the rules cost, the aux summary, outcomes per action

← Reports

Operations report

PDF | Excel 4 | Send by email

The handlings in the period rolled up: how much was handled, what it was predicted to cost, what obeying the rules cost, and — once outcomes are recorded — what it actually cost. The figures are computed from the record; a written summary is added only when the auxiliary model is available.

Last 30 days | Last 90 days | All time — every handling on record

Handlings	Actions raised	Committed (predicted)	Realized vs predicted
2	3	Rp 123,000,000	+Rp 2,500,000
2 done	3 sent · 0 awaiting approval	1 decided on trusted evidence · 1 with a caveat	1 comparable action(s)

Summary

Phrased by the auxiliary model from the figures above. It was given no number to invent.

Across every handling on record, 2 runs were completed covering 2 disruptions, with 3 actions raised, 3 sent, and 0 pending. Predicted total cost was Rp 123,000,000, and for the 1 comparable settled action the actual cost was Rp 24,500,000 versus a predicted Rp 22,000,000, a delta of Rp 2,500,000. Standing exposure remains from 1 action still owing an outcome and 1 recorded outcome that carries no figure; in addition, obeying the rules cost Rp 46,000,000 across 1 disruption, with rules verified in 0 of 4 cases and 1 trusted-evidence decision carrying a caveat.

What the rules cost

Where a rule struck out a cheaper plan, the difference between the plan that was taken and that one. The price of the rule, not an overrun.

Cost of compliance	Chosen plans	Cheaper, struck out
Rp 46,000,000	Rp 116,000,000	Rp 70,000,000
across 1 disruption(s)	what the compliant plans were predicted to cost	what they would have cost if no rule applied

Handling	Chosen	Cheaper, struck out	Cost of the rule	Struck out by
Typhoon closes Ningbo port 27 Oct 2026	C, B Rp 116,000,000	C, E Rp 70,000,000	Rp 46,000,000	E TKDN drops to 34.8%, contract floor 40%

Handlings (2)

Handling	Kind	Status	Started	Steps	Actions	Model time
Typhoon closes Ningbo port	port_closure	done	27 Oct 2026 21:15 WIB	1	3	-
Supplier halt threatens TG0011 at plant 1010	supplier_disruption	done	27 Oct 2026 21:14 WIB	6	0	-

Actions (3)

1 of these went out and still owe an outcome. Until those are recorded the realized side below is incomplete.

Action	What it does	Handling	Value	Status	What happened
stock_transfer own authority	M-4471 · KRW1 → SBY1 · qty 5.0	Typhoon closes Ningbo port	Rp 7,000,000	sent	owes an outcome
stock_transfer own authority	M-4471 · SBY1 → KRW1 · qty 22.0	Typhoon closes Ningbo port	Rp 22,000,000	sent	delivered Rp 24,500,000
draft_po own authority	M-4471 · supplier SUP-GRESIK · qty 30.0	Typhoon closes Ningbo port	Rp 94,000,000	sent	late no figure

Realized vs predicted

Predicted	Actual	Delta
Rp 22,000,000	Rp 24,500,000	+Rp 2,500,000
1 comparable action(s)		came in over prediction

Drawn from 1 action(s) that have both a prediction and a realized figure. A further 1 recorded outcome(s) carry no figure and are not in this comparison. 1 action(s) that went out still owe an outcome.

The rules behind these decisions

A rule counts as verified only where the pack states it, cites the document it was read from, and a person has checked it there.

0 of 4 rules verified

Assumptions, applied and labelled as such: tkdn_contract_floor_pct, tkdn_current_pct, lartas_new_origin_working_days, default_port_dwell_days

Rule pack: /Users/lana/Desktop/lam/socrates/sigap/references/rule-pack.json

The operations rollup: period, honest delta, cost of compliance, rule trust

done supplier_disruption

PDF | Excel 4 | Send by email | Markdown | Copy

Supplier halt threatens TG0011 at plant 1010

Supplier notice: production halted, deliveries of TG0011 to plant 1010 suspended

Started	Took	Steps	Tool calls	Actions	Model time
27 Oct 2026 21:14 W…	0m 57s	6	3	0	-

Summary

Phrased by the auxiliary model from the figures in this report. It was given no number to invent.

A supplier halt was reported for TG0011 to plant 1010, with deliveries suspended. It was concluded that there is no evidence of an operational shortage at plant 1010 and no customer commitments at risk were established, while inbound purchase orders also could not be confirmed.

Actions (0)

None were raised.

In the Excel file (4 sheets)

Summary	13 rows
Steps	6 rows
get_open_purchase_orders	3 rows
get_material_stock	1 rows

Every table the handling produced: its actions and steps, any table in the decision, and each row a tool read, one sheet per tool with the step and source beside it.

Related

The decision page →

The trace →

Decision

What happened and what it costs

A supplier halt was reported for **TG0011** to **plant 1010**.

Based on the impact check, I do **not** have evidence of an operational shortage at plant 1010:

- Stock at **1010**: **10,020 PC unrestricted**
- Additional stock not currently free to use:
  - **511 PC** in quality inspection
  - **165 PC** blocked
- Consumption for TG0011 at 1010: **0.0 PC/day** (**derived** data)
- BOM trace: **no finished goods found consuming TG0011** (**live** data)
- Customer commitments at risk: **none established**

I also could not confirm any inbound purchase orders for **plant 1010** tied to the failed supplier. The open purchase orders found for TG0011 were for **plant 1710**, not 1010.

Cost: no cost figure is available, and I will not emit one. There is no established customer impact, and the critical supplier identity for the failed source is **missing**.

Which options were rejected, and why

I did not take sourcing, logistics, compliance, or costing further, because the evidence was already enough to decide there is **no current confirmed service risk** at plant 1010.

Implicitly rejected for now:

- **Emergency replacement sourcing**
  - Not justified yet: no shortage or customer exposure established.
- **Expedite / logistics intervention**
  - Not justified yet: no confirmed inbound to plant 1010 at risk was found.
- **Reallocation / prioritization**
  - Not justified yet: no consuming product or customer commitment exposure was found.

Recommendation

Recommendation: monitor only; do not intervene yet.

Why this beats the nearest alternative:

- The nearest alternative is to start emergency sourcing or expedites immediately.
- That would add effort and likely cost without evidence of a shortage, demand, or customer exposure.
- Current evidence shows **ample unrestricted stock** and **no traceable downstream usage** for TG0011 at plant

The live agent run's own report

← Log

done supplier_disruption run_9f74020e569a

Trace · Supplier halt threatens TG0011 at plant 1010

Supplier notice: production halted, deliveries of TG0011 to plant 1010 suspended

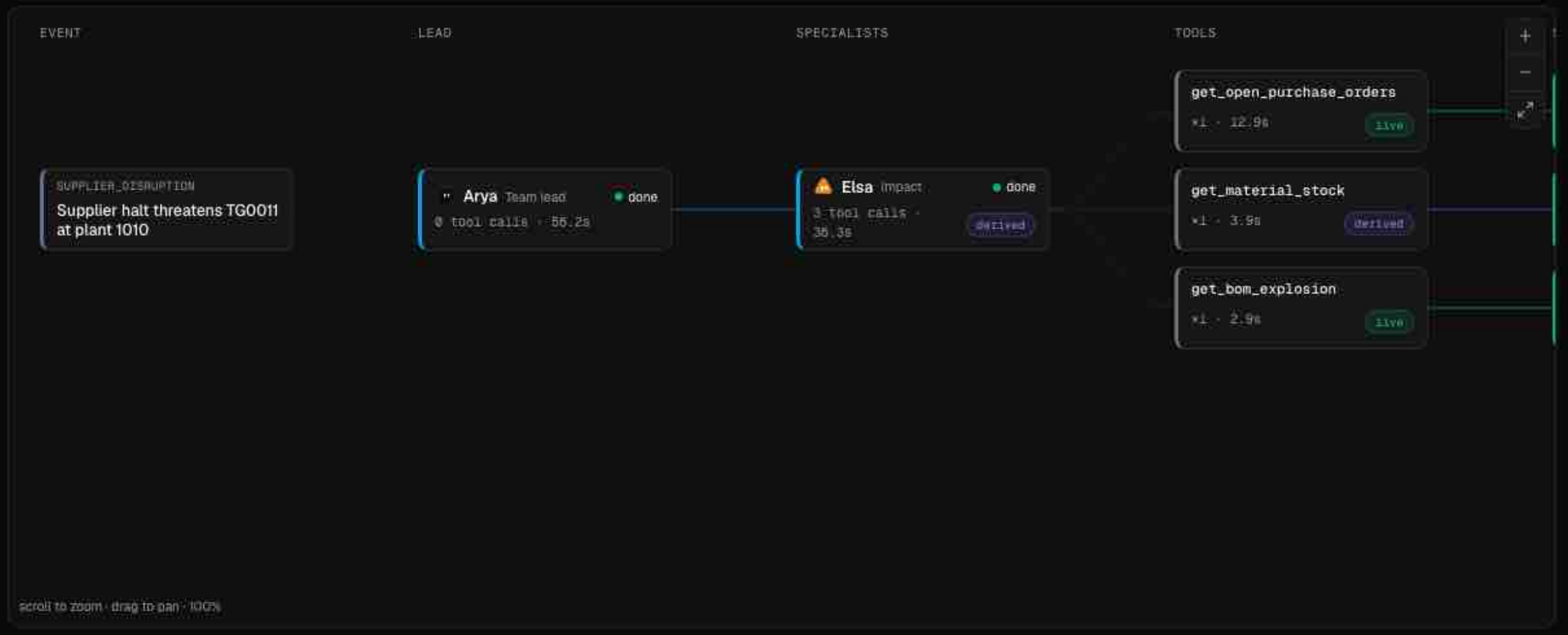
Started	Took	Steps	Tool calls	Agents · sources	Model time
27/10/2026, 21:14:36	57s	6	3	2 · 3	-

What the tool calls read: live · 2 derived · 1

The decision for this handling

Topology

Who asked whom, which tool each called, and where every figure was read from. Click a node to open its details: its numbers, what it connects to, and every step through it.



SAP · upload · fixture / profile · outside feed · engine (calculation) · SIGAP record · action · a read edge is coloured by the origin of what came back · ×n is how many calls

Every step, in order

All | Tool calls | Delegations | Findings | live 2 | derived 1 | Search tool, material, note... | 6 of 6 steps

#1	21:14:45	Arya → Elsa delegates	A supplier disruption has halted deliveries of material TG0011 to plant 1010. Trace the impact for this material at plant 1010: current available stock, inbound su...	
#2	21:15:01	Elsa calls	get_open_purchase_orders(material=TG0011)	3 rows 12932 ms live
#3	21:15:05	Elsa calls	get_material_stock(material=TG0011, plant=1010)	1 rows 3947 ms derived
#4	21:15:08	Elsa calls	get_bom_explosion(material=TG0011)	0 rows 2892 ms live
#5	21:15:21	Elsa reports	Here's what I could establish for the disruption affecting material **TG0011** at **plant 1010**.	
#6	21:15:33	Arya answers	## What happened and what it costs	

The trace of the live run: every step and every tool call

Typhoon closes Ningbo port

Maritime advisory: CNNGB closed, vessels held at anchorage

Kind	Status	Model cost
port_closure	done	Rp 0
Started	**Finished**	**Steps · tool calls**
27 Oct 2026 21:15 WIB	27 Oct 2026 21:15 WIB	1 · 1

Summary

A typhoon closed Ningbo port, with CNNGB closed and vessels held at anchorage, disrupting supply. The decision was to recommend C+B, using an internal transfer plus Gresik, because E was blocked on TKDN; the stock transfer was delivered, the draft PO was late, and the rules added Rp 46,000,000 by forcing C, B at Rp 116,000,000 instead of C, E at Rp 70,000,000. Residual risk remains because the TKDN basis is not yet trustworthy: the contract floor of 40% versus current 34.8% depends on assumptions rather than verified rules in compliance.check_local_constraints.

Phrased by the auxiliary model from the figures below; it was given no number to invent.

Decision

Recommend C+B (internal transfer plus Gresik). E blocked on TKDN.

> Not yet trustworthy: compliance.check_local_constraints

> 2 of the 2 rules applied here are not verified against a source, so they are assumptions written down rather than rules: tkdn_contract_floor_pct, tkdn_current_pct

What the rules cost

Obeying the rules on this disruption cost Rp 46,000,000: the plan that was chosen, set against the cheaper one a rule struck out.

	Plan	Cost
Chosen	C, B	Rp 116,000,000
Cheaper, struck out	C, E	Rp 70,000,000
Difference	what the rules cost	Rp 46,000,000

Struck out	Why
E China spot purchase	TKDN drops to 34.8%, contract floor 40%

This is the price of the rule, not an overrun. It is on the record so the decision can be defended without re-deriving it.

Outcome: realized vs predicted

Action	Predicted	Actual	Delta
stock_transfer (delivered)	Rp 22,000,000	Rp 24,500,000	Rp 2,500,000
draft_po (late)	Rp 94,000,000	-	-

1 of 2 recorded outcome(s) carry no realized figure, so no delta is stated for them. A dash is 'not known', never zero.

Generated PDF (handling)

SIGAP · EVERY HANDLING ROLLED UP

Operations report

every handling on record. How much was handled, what it was predicted to cost and, once outcomes are recorded, what it actually cost.

Period
every handling on record

Handlings
2 (2 done)

Actions raised
3 (3 sent, 0 awaiting)

Committed (predicted)
Rp 123,000,000

Realized vs predicted
+Rp 2,500,000 (1 comparable)

Outcomes still owed
1

What the rules cost
Rp 46,000,000

Rules verified
0 of 4

Decided on trusted evidence
1

Decided with a caveat
1

Summary

Across every handling on record, 2 runs were completed covering 2 disruptions, with 3 actions raised, 3 sent, and 0 pending. Predicted total cost was Rp 123,000,000; for the 1 comparable settled action, actual cost was Rp 24,500,000 against a predicted Rp 22,000,000, a delta of Rp 2,500,000. Standing exposure remains with 1 action still owing an outcome, 1 recorded outcome carrying no figure, and rules verified at 0 of 4.

Handlings (2)

Handling	Kind	Status	Started	Steps	Actions	Model cost
Typhoon closes Ningbo port	port_closure	done	27 Oct 2026 21:15 WIB	1	3	Rp 0
Supplier halt threatens TG0011 at plant 1010	supplier_disruption	done	27 Oct 2026 21:14 WIB	6	0	Rp 0

Actions (3)

Handling	Action	What it does	Value	Status
Typhoon closes Ningbo port	stock_transfer	M-4471 · KRW1 -> SBY1 · qty 5.0	Rp 7,000,000	sent
Typhoon closes Ningbo port	stock_transfer	M-4471 · SBY1 -> KRW1 · qty 22.0	Rp 22,000,000	sent
Typhoon closes Ningbo port	draft_po	M-4471 · supplier SUP-GRESIK · qty 30.0	Rp 94,000,000	sent

Realized vs predicted

	Predicted	Actual	Delta
1 comparable action(s)	Rp 22,000,000	Rp 24,500,000	Rp 2,500,000

Across 1 comparable action(s), realized cost came in over prediction by Rp 2,500,000. A further 1 recorded outcome(s) carry no figure and are not in this comparison. 1 action(s) that went out still owe an outcome.

Generated PDF (operations report)

Supplier halt threatens TG0011 at plant 1010

Supplier notice: production halted, deliveries of TG0011 to plant 1010 suspended

Kind	Status	Model cost
supplier_disruption	done	Rp 0
Started	**Finished**	**Steps · tool calls**
27 Oct 2026 21:14 WIB	27 Oct 2026 21:15 WIB	6 · 3

Summary

A supplier halt was reported for TG0011 to plant 1010, with deliveries suspended. The review found no evidence of an operational shortage at plant 1010, no finished goods consuming TG0011, no customer commitments at risk, and no confirmed inbound purchase orders.

Phrased by the auxiliary model from the figures below; it was given no number to invent.

Decision

What happened and what it costs

A supplier halt was reported for **TG0011** to **plant 1010**.

Based on the impact check, I do **not** have evidence of an operational shortage at plant 1010:

- Stock at **1010**: **10,020 PC unrestricted**
- Additional stock not currently free to use:
- **511 PC** in quality inspection
- **165 PC** blocked
- Consumption for TG0011 at 1010: **0.0 PC/day** (**derived** data)
- BOM trace: **no finished goods found consuming TG0011** (**live** data)
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I also could not confirm any inbound purchase orders for **plant 1010** tied to the failed supplier. The open purchase orders found for TG0011 were for **plant 1710**, not 1010.

Cost: no cost figure is available, and I will not emit one. There is no established customer impact, and the critical supplier identity for the failed source is **missing**.

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I did not take sourcing, logistics, compliance, or costing further, because the evidence was already enough to decide there is **no current confirmed service risk** at plant 1010.

Implicitly rejected for now:

- **Emergency replacement sourcing**
- Not justified yet: no shortage or customer exposure established.
- **Expedite / logistics intervention**
- Not justified yet: no confirmed inbound to plant 1010 at risk was found.
- **Reallocation / prioritization**
- Not justified yet: no consuming product or customer commitment exposure was found.

Generated PDF of the live agent run